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Colonization of Rakata (Krakatau Is.) by non-migrant land birds from 1883 to 1992 and implications for the value of island equilibrium theory

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Abstract. Colonization curves and changes in immigration and extinction rates of non-migrant land birds since the 1883 eruption are provided for the recolonization of Rakata (Krakatau Is.), including results of surveys made in 1990 and 1992. The contention by Bush & Whittaker (1991) that non-monotonic changes in observed immigration and extinction rates of birds and butterflies cast doubt on the ability of the equilibrium theory of island biogeography (MacArthur & Wilson, 1967) to provide the basis for a predictive recolonization model is examined in the light of our recent data.

We conclude that rates of immigration and extinction are highly sensitive to decisions as to immigrant status or extinction of species, particularly where intersurvey intervals are short and turnover numbers are low. We note that for no group of organisms on the island is there an equilibrium in number of species, most of the observed turnover having been successional. Contrary to Bush & Whittaker (1991), the number of non-migrant land bird species has not declined since 1951 and numbers are still rising, albeit more slowly than in the early decades of this century. The colonization curve is flattening, and immigration rate is falling towards extinction rate, in general although not precise accord with the model.

On comparing the colonization dynamics of birds with those of plants and butterflies and mindful of the qualifications MacArthur & Wilson (1967) placed on the applicability of their model to cases where succession is important in the early stages of colonization, we conclude that it is premature to discard their dynamic approach as a theoretical framework for the study of animal colonization of the Krakataus.

Key words. Krakatau, island biogeography, immigration and extinction curves, resident land birds.

INTRODUCTION

One conclusion of a recent paper by Bush & Whittaker (1991) on the colonization of the Krakatau islands since the 1883 eruption was that changes in observed immigration and extinction rates on the island of Rakata (Krakatau's remnant) are non-monotonic and that this casts doubt on the ability of equilibrium theory (MacArthur & Wilson, 1967) to provide the basis for a predictive recolonization model. Their conclusions were partly based on the data set for resident land birds (excluding owls), including a 1989 survey by Haag & Bush (1990) but excluding data from our surveys in 1984 and 1985 (Thornton, *et al.*, 1990b; Zann *et al.*, 1990a,b). In a recent short note Bush & Whittaker (1991) observed that the census data provided by Zann *et al.* (1990b) support the non-monotonicity of the rate curves, and thus their argument that the pattern is 'therefore contra to MacArthur & Wilson's (1967) model of island colonization.' We have since made surveys in 1990 and 1992 and in this paper we consider the effect of our 1984, 1985, 1990 amd 1992 data on Bush & Whittaker's conclusions, specifically concerning the colonization dynamics of birds and, more generally, concerning the value of the dynamic island biogeographical approach to studies of the colonization of the Krakataus.

In September 1984 R. A. Zann, A. S. Adikerana, M. V. Walker and G. W. Davison (the last three being ornithologists with considerable experience in the region), and again in August 1985 (Zann), surveyed birds on Rakata at three sites in coastal forest (73 man-hours visual, 28 man-hours spotlighting, $3446 \times m^2$ rain-free hours mist-netting, 2 hours sound recording), at eight sites in *Neonauclea* forest (79, 112, 2584 and 0.3 respectively), at five sites in *Neonauclea-Ficus* moss forest (23 man-hours visual) and at one site in *Schefflera-Leucosyke* summit scrub (10 man-hours visual) (Zann *et al.* 1990*b*). The 1986 survey did not include Rakata. Further ornithological surveys of the island were made in August–September 1990 by Zann and van Balen (another ornithologist with considerable previous experience in the region), and in July 1992 by van Balen. Rakata was surveyed on 31 August, 1 September and 2 September 1990 and on 6 July 1992. In 1990 observations began at 0500 hours each day and totalled over 50 man-hours of visual survey and sound recording; in 1992 there were 10 man-hours of sound and visual survey and $3824m^2 \times$ rain-free hours of mist-netting.

The Data Set

Like Bush & Whittaker (1991), in previous papers on bird colonization of the Krakataus our group has confined its attention to non-migrant land birds (excluding shore birds). We list and number these species (Tables 1 and 2) in the same order as in Table 2 of Thornton *et al.* (1990b). Species are counted as absent if they were not recorded from two successive surveys of suitable habitat. A species is therefore assessed as having been present, although not recorded in a survey (indicated by an asterisk in the tables), if it was recorded in the immediately preceding and subsequent surveys, unless there are good reasons, such as conspicuousness by sight or sound of the species concerned, to decide otherwise. The assessment is thus made on the basis of minimum turnover and is more conservative than the unassessed records.

The data from our 1990 and 1992 surveys necessitates modification of our previous species list (Thornton *et al.* 1990b), and the incorporation of these data and those of our 1984 and 1985 surveys requires changes to the list provided by Bush & Whittaker (1991). These changes are detailed below.

Species recorded as present on Rakata prior to the 1980s that were not so listed by Bush & Whittaker (1991)

Crested goshawk, *Accipiter trivirgatus* (15 in Table 2). Hoogerwerf (1953) believed this was probably misidentified by Dammerman's group (Chasen, 1937; Dammerman, 1948) as the smaller sparrowhawk, *Accipiter virgatus.* The sparrowhawk was regarded by Dammerman (1948) and Hoogerwerf (1948, 1953) as comprising two subspecies: *virgatus*, resident on Java, and *gularis*, which is migratory and breeds in NE Asia. King, Woodcock & Dickinson (1975) and MacKinnon (1988) recognised these as separate species, the resident besra (*A. virgatus*) and the Japanese sparrowhawk (*A. gularis*). Chasen (1937) and Dammerman (1948) recorded a species of *Accipiter*, which they regarded as probably being the migratory *gularis*, from Rakata in 1919 (and also from Sertung in 1933). Hoogerwerf (1953) noted that the species recorded by Chasen and Dammerman was seen hovering and flying in circles and believed this behaviour to indicate a misidentification of the larger, resident, crested goshawk, *Accipiter trivirgatus.* Hoogerwerf was aware of the differences between these species for in the same paper he recorded a small *Accipiter* from Sertung that he stated was probably the besra. Although there is doubt about the identity of the sightings by Chasen and Dammerman we have opted to accept Hoogerwerf's view and list the goshawk (*A. trivirgatus*) rather than the Japanese sparrowhawk (*A. gularis*) as present on Rakata in 1919, on Rakata and Sertung in 1933 (Dammerman 1948) and on Sertung in 1951 (Hoogerwerf, 1953). The species has not been recorded since 1951.

Koel, *Eudynamys scolopacea* (17 in Tables 1 and 2). This species was fairly numerous on Rakata and Sertung in 1919, less so in 1933, and was recorded on Rakata in 1951 by Hoogerwerf (1953), who stated 'we possibly heard the species once (on 6 October)', but nevertheless listed the species as present (see also below).

Brahminy kite, *Haliastur indus* (21). The species was recorded as present on Rakata in 1919 (perhaps one individual) and 1933 (and on Sertung in 1920, 1932 and 1951) (Bartels, 1920; Dammerman, 1922, 1929, 1948; Chasen, 1937; Hoogerwerf, 1953). See also below.

Serpent eagle, *Spilornis cheela* (35). This eagle was heard frequently and seen on Rakata in 1951 by Hoogerwerf (1953). See also below.

Species recorded on Rakata in the 1980s, 1990 or 1992 that were not recorded as present on the island by Bush & Whittaker for the period 1983–1989

Large-billed crow, *Corvus macrorhynchos* (4). See also below.

Pink-necked pigeon, *Treron vernans* (6). Previously fairly common on Rakata, but not found by Hoogerwerf in 1951 (it was present on Sertung in 1952), this pigeon was recorded on Rakata in 1984, 1985, 1989 and 1990 (Thornton *et al.*, 1990b; Zann *et al.*, 1990b; Haag & Bush, 1990; and our 1990 survey). In our 1992 survey it was very vocal and assessed as frequent (Table 1). It is counted as present on Rakata. (The species was found on Anak Krakatau during our 1986 expedition).

Savannah nightjar, *Caprimulgus affinis* (7). Heard on the island (Zwarte Hoek) in 1985 (Zann *et al.*, 1990b). See also below.

Green-winged pigeon, *Chalcophaps indica* (8). Formerly very common on Rakata, and previously recorded as breeding there (Dammerman, 1948), several individuals (of both sexes) were netted on the island in 1984 and 1985 (Zann *et al.*, 1990b) and one in 1992. Although it was not recorded on Rakata in 1989 or 1990, our 1990 survey did not include mist-netting in the *Barringtonia* formation, the method of discovery of this secretive bird in 1984, 1985 and 1992. It is counted as present on Rakata.

Edible-nest swiftlet, *Aerodramus fuciphagus* (9). This species was seen on Rakata in 1984 (Zann *et al.*, 1990b) and was assessed as frequent there in 1990 (Table 1).

Koel, *Eudynamys scolopacea* (17). Recorded on Rakata in our 1990 survey and assessed as being frequent on the island, we believe the koel is likely to have been present since 1951 but missed in the 1980s surveys (see above). It was discovered on Anak Krakatau in 1992. The koel's calls are 'not heard for months on end' and it is 'rarely seen' (Hoogerwerf, 1953) so that, for this bird in particular, absence of evidence is not necessarily evidence of absence.

Brahminy kite, *Haliastur indus* (21). This species was seen only once in the surveys of the last decade (in 1984), soaring over Rakata's Zwarte Hoek (Zann *et al.*, 1990b). We previously believed the species may have been missed in 1985 (Thornton *et al.*, 1990b), but it has not been recorded in any subsequent survey (Table 1) and its status on the island must be regarded as doubtful. We have not counted it on assessment (see also below).

Pied triller, *Lalage nigra* (24). Both sexes were twice

TABLE 1. Resident land birds recorded on the Krakataus in the 1980s and their frequency on individual islands in our surveys of August 1990 and July 1992. Species listed and numbered in same order as Table 2 of Thornton *et al.* (1990b). Rakata, Sertung, Panjang and Anak Krakatau denoted by initial letters. Rakata in bold where presence not in doubt.

Species	1983	1984	1985	1989	1990				1992			
					R	S	P	A	**R**	S	P	A
4. *Corvus macrorhynchos*	(R)A	A	—	—	0	0	0	0	0	0	0	0
5. *Centropus bengalensis*$^{\phi}$	+	SA	A	A	0	0	0	3	0	0	0	3
6. *Treron vernans*$^{\blacktriangle}$	*	**RS**	**RSP**	**RS**	**3**	3	0	0	**3**	0	2	0
7. *Caprimulgus affinis*	—	A	(R)(S) (P)A	A	0	0	0	4	0	0	0	3
8. *Chalcophaps indica*	*	**RSP**	**RSP**	*S	*0	4	4	0	1	2	0	0
9. *Aerodramus fuciphagus*	*	**RSPA**	*PA	*	**3**	4	3	3	*0	2	3	3
10. *Halcyon chloris*	**RSA**	**RSPA**	**RSPA**	**RSPA**	**5**	4	4	4	**4**	4	4	4
11. *Pycnonotus goiavier*	*A	**RSPA**	**RSPA**	**RSPA**	**5**	5	5	5	**4**	5	4	4
12. *Oriolus chinensis*	*A	**RSPA**	**RSPA**	**RSA**	**4**	4	4	3	**3**	3	4	3
13. *Artamus leucorhynchus*	*	**RSPA**	**RSPA**	**RSPA**	**3**	5	3	4	**3**	4	3	4
17. *Eudynamys scolopacea*	*	*	*	*	**3**	3	2	0	*0	0	3	1
18. *Ducula bicolor*	**RS**	**RS**	**RSP**	**R**	0	4	?1	0	0	4	3	0
19. *Hirundo tahitica*	*	**RSA**	**RA**	*SP	**3**	3	0	2	*0	2	0	1
20. *Amaurornis phoenicurus*	—	A	A	S	0	1	0	2	0	0	0	2
21. *Haliastur indus*	—	(R)	—	—	0	0	0	0	0	0	0	0
22. *Haliaetus leucogaster*	**RSP**	**RSP**	**RSA**	**RSPA**	**2**	3	2	2	**2**	2	2	2
23. *Picoides moluccensis*	*	*S	*SP	**RSP**	*0	4	3	0	**3**	1	3	0
24. *Lalage nigra*	*	*	**RP**	*A	**1**	3	0	2	**3**	2	0	3
25. *Copsychus saularis*	**RS**	**RSPA**	*SPA	**RSPA**	**3**	4	4	5	**4**	4	4	4
26. *Pachycephala cinerea*	*	**RSPA**	**RSPA**	**RSPA**	**4**	4	4	5	**4**	4	4	5
27. *Aplonis panayensis*	**R**	**RPA**	**RSP**	**RSA**	**4**	5	4	0	**3**	4	4	0
28. *Anthreptes malacensis*	**R**	**RSPA**	*SP	**RSP**	**4**	5	4	5	**4**	4	3	3
29. *Nectarinia jugularis*	*	**RSPA**	*SPA	**RSPA**	*0	3	3	5	*0	3	0	5
30. *Dicaeum trigonostigma*	*	**R**	**R**	*	**4**	?	3	0	**4**	0	3	0
31. *Macroypygia phasianella*	**RS**	**RS**	**RSPA**	**RS**	**4**	4	4	1	**4**	0	3	0
32. *Collocalia esculenta*	*	*	**RP**	SA	0	0	0	1	0	0	0	0
33. *Gerygone sulphurea*	*	**RSPA**	*SPA	**RSPA**	*0	3	4	5	**3**	3	3	4
34. *Cyornis rufigastra*	**RS**	**RSPA**	**RSPA**	**RSPA**	**4**	4	4	4	**4**	4	5	4
35. *Spilornis cheela*	—	—	?R	—	?3	2	?1	0	0	0	0	0
37. *Ptilinopus melanospila*	*	**R**	**RP**	*S	**5**	5	4	1	**4**	4	4	1
38. *Aethopyga siparaja***	**R**	**RP**	**RP**	*	**3**	?	3	0	**3**	0	0	0
39. *Pynonotus plumosus*	*	**RS**	—	—	0	0	0	0	0	0	0	0
40. *Falco severus*	—	—	—	A	0	0	0	0	0	0	0	0
41. *Ictinaëtus malayensis*	**R**	**RP**	**RSP**	**R**	?	0	2	1	0	0	0	0
42. *Ducula aena*	**RSP**	**RP**	*P	**RSP**	**3**	4	4	0	**4**	2	3	0
43. *Tyto alba*	(R)***	A	SA	—	0	0	0	3	0	0	0	3
44. *Apus affinis*	*	**RSPA**	**RSA**	**RPA**	**1**	0	0	0	**3**	0	0	3
45. *Zoothera interpres*	*	**R**	*(A)$^{\blacktriangle}$	*	**3**	0	0	0	*0	3	0	0
46. *Corvus splendens*	—	—	*(A)$^{\blacktriangle}$	—	0	0	0	0	0	0	0	0
47. *Spizaetus cirrhatus*	—	—	—	(A)	**3**	2	1	0	**2**	0	2	0
48. *Pernis ptilorhynchus*	—	—	—	(R)	0	0	0	0	0	0	0	0
49. *Falco peregrinus*	—	—	—	—	0	0	0	1	**1**	0	0	2
50. *Zosterops palpebrosus****	—	—	—	—	0	0	0	1	0	0	0	0
51. *Pericrocotus cinnamomeus*	—	—	—	—	0	0	0	0	0	1	0	0
52. *Lonchura leucogastroides*	—	—	—	—	0	0	0	0	0	0	0	1
53. *Lonchura punctulata*	—	—	—	—	0	0	0	0	0	0	0	1
54. *Treron griseicauda*	—	—	—	—	0	0	0	0	**1**	0	0	0

*Assumed present on Rakata; **previously recorded as *Aethopyga mystacalis*; *** in 1982; ( ) single observation, presumed straggler, not accepted as resident on assessment; ? record questionable, requires confirmation, not accepted on assessment; 1976 record on Anak Krakatau (B. King, *pers. comm.*); + assumed present on archipelago; $^{\phi}$ recorded on Panjang in 1982 by Ibkar Kramadibrata *et al.* (1986); $^{\blacktriangle}$ present on Anak Krakatau in 1986. 1990 and 1992 frequencies: 0 absent, 1 rare (1 observation), 2 occasional, 3 frequent, 4 common, 5 very common. 1983 records from Bush & Newsome (1986) and Bush *in litt*; 1984 and 1985 records from Thornton *et al.* (1990a), Zann *et al.*, (1990a,b) 1989 records from Haag & Bush (1990).

Note:

50–53 are new records not involving Rakata, as follows.

50. Oriental whiteye, *Zosterops palpebrosus.* One individual was seen on Anak Krakatau by Mr Torben Lund, a Danish birdwatcher, on 29 August 1990. Not counted as resident.
51. Small minivet, *Pericrocotus cinnamomeus.* A single female was seen and heard by van Balen in July 1992 on the Sertung spit. This is a rather common species on Java and also is a resident of Panaitan I. Not counted as resident.
52. Javan white-bellied munia, *Lonchura leucogastroides.* One individual was mist-netted in July 1992 on the E Foreland of Anak Krakatau. Not counted as resident.
53. Scaly-breasted munia, *Lonchura punctulata.* A small flock was heard by van Balen several times on the E. Foreland of Anak Krakatau in July 1992, and identified from the unmistakable ‘tepee’ or ‘kidee’ call. This is the only munia resident on Sebesi I. Not counted as resident on the Krakataus.

TABLE 2. Resident land birds of the Krakataus, 1883–1992. For symbols see Table 1. Data from Thornton *et al.* (1990b) and Table 1. 1983–1992 column pools all the data of Table 1. The two preceding columns (in square brackets) show these data as two subsets reflecting the two main periods of sampling.

	1908	1919–21	1932–34	1951–52	[1983–85	1989–92]	1983–1992
1. *Alcedo caerulescens*	**R**	—	—	—	[—	—]	—
2. *Pycnonotus aurigaster*	**R**	—	—	—	[—	—]	—
3. *Lanius schach*	**R**	**RS**	—	—	[—	—]	—
4. *Corvus macrorhynchos*	**R**P	**RS**	**RS**P	S	[(R)A	—]	(R)A
5. *Centropus bengalensis*	**R**	**RS**	**RS**	**R**	[SP$^{\phi}$A	A]	SPA
6. *Treron vernans*	**R**	**RS**	**RS**P	*S	[**R**SP	**R**SP]	**R**SP
7. *Caprimulgus affinis*	**R**	**RS**	**RS**	A	[(R)(S) (P)*A*	A]	(R)(S) (P)A
8. *Chalcophaps indica*	**R**	**RS**	**RS**	**R**	[**R**SP	**R**SP]	**R**SP
9. *Aerodramus fuciphagus*	**R**P	**RS**	**RS**	**R**	[**R**SPA	**R**SPA]	**R**SPA
10. *Halcyon chloris*	**R**	**RS**	**RS**P	**R**	[**R**SPA	**R**SPA]	**R**SPA
11. *Pycnonotus goiavier*	**R**	**RS**	**RS**P	**R**A	[**R**SPA	**R**SPA]	**R**SPA
12. *Oriolus chinensis*	**R**SP	**RS**	**RS**P	**R**	[**R**SPA	**R**SPA]	**R**SPA
13. *Artamus leucorhynchus*	**R**P	**RS**	**RS**	**R**	[**R**SPA	**R**SPA]	**R**SPA
14. *Centropus sinensis*	+	S	—	—	[—	—]	—
15. *Accipiter trivirgatus*	*	**R**	**RS**	S	[—	—]	—
16. *Geopelia striata*	*	**R**	+	S	[—	—]	—
17. *Eudynamys scolopacea*	—	**RS**	**RS**P	**R**	[*	**R**SPA]	**R**SPA
18. *Ducula bicolor*	—	**RS**	**RS**	—	[**R**SP	**R**SP]	**R**SP
19. *Hirundo tahitica*	—	**RS**	**RS**	*	[**R**SA	**R**SPA]	**R**SPA
20. *Amaurornis phoenicurus*	—	**RS**	**RS**P	**RS**	[A	SA]	SA
21. *Haliastur indus*	—	**RS**	**RS**	S	[(R)	—]	(R)
22. *Haliaeetus leucogaster*	—	**RS**	**RS**P	**RS**A	[**R**SPA	**R**SPA]	**R**SPA
23. *Picoides moluccensis*	—	**R**	**RS**	**R**	[*SP	**R**SP]	**R**SP
24. *Lalage nigra*	—	**RS**	**RS**	**R**	[**R**P	**R**SA]	**R**SPA
25. *Copsychus saularis*	—	**RS**	**RS**	**R**	[**R**SPA	**R**SPA]	**R**SPA
26. *Pachycephala cinerea*	—	**RS**	**RS**	**R**	[**R**SPA	**R**SPA]	**R**SPA
27. *Aplonis panayensis*	—	**RS**	**RS**P	**R**	[**R**SPA	**R**SPA]	**R**SPA
28. *Anthreptes malacensis*	—	**RS**	**RS**	**R**	[**R**SPA	**R**SPA]	**R**SPA
29. *Nectarinia jugularis*	—	**RS**	**RS**P	**R**	[**R**SPA	**R**SPA]	**R**SPA
30. *Dicaeum trigonostigma*	—	**RS**	**RS**	**R**	[**R**	**R**?SP]	**R**?SP
31. *Macropygia phasianella*	—	—	S	**R**	[**R**SPA	**R**SPA]	**R**SPA
32. *Collocalia esculenta*	—	—	**R**	**R**	[**R**P	*SA]	**R**SPA
33 *Gerygone sulphurea*	—	—	**RS**P	**R**	[**R**SPA	**R**SPA]	**R**SPA
34 *Cyornis rufigastra*	—	—	**RS**P	**R**	[**R**SPA	**R**SPA]	**R**SPA
35 *Spilornis cheela*	—	—	—	**R**	[?R	?RS?P]	?RS?P
36. *Rhaphidura leucopygialis*	—	—	—	**R**	[—	—]	—
37. *Ptilinopus melanospila*	—	—	—	**R**	[**R**P	**R**SPA]	**R**SPA
38. *Aethopyga siparaja***	—	—	—	**RS**	[**R**P	**R**?SP]	**R**?SP
39. *Pycnonotus plumosus*	—	—	—	**RS**	[**R**S	—]	**R**S
40. *Falco severus*	—	—	—	S	[—	A]	A
41. *Ictinaëtus malayensis*	—	—	—	—	[**R**SP	**R**P(A)]	**R**SP(A)
42. *Ducula aena*	—	—	—	—	[**R**SP	**R**SP]	**R**SP
43. *Tyto alba*	—	—	—	—	[SA	A]	SA
44. *Apus affinis*	—	—	—	—	[**R**SPA	**R**PA]	**R**SPA
45. *Zoothera interpres*	—	—	—	—	[**R**(A)$^{\blacktriangle}$	**R**S]	**R**S(A)$^{\blacktriangle}$
46. *Corvus splendens*	—	—	—	—	[(A)$^{\blacktriangle}$	—]	(A)$^{\blacktriangle}$
47. *Spizaetus cirrhatus*	—	—	—	—	[—	**R**SP(A)]	**R**SP(A)
48. *Pernis ptilorhynchus*	—	—	—	—	[—	(R)]	(R)
49. *Falco peregrinus*	—	—	—	—	[—	(R)A]	(R)A
50. *Zosterops palpebrosus*	—	—	—	—	[—	(A)]	(A)
51. *Pericrocrotus cinnamomeus*	—	—	—	—	[—	(S)]	(S)
52. *Lonchura leucogastroides*	—	—	—	—	[—	(A)]	(A)
53. *Lonchura punctulata*	—	—	—	—	[—	(A)]	(A)
54. *Treron griseicauda*	—	—	—	—	[—	(R)]	(R)

seen on Rakata in 1985 (Zann *et al.*, 1990b) and the species was again seen there in 1990 and 1992. It is counted as present on Rakata.

White-bellied swiftlet, *Collocalia esculenta* (32). Although we recorded this species on Rakata (and Panjang) in 1985, we found it only on Anak Krakatau in 1990 and did not record it in 1992 (Table 1). Bush & Whittaker (1991) recorded it as present on Sertung and Anak Krakatau in 1989. We assess it as absent now from Rakata.

Serpent eagle, *Spilornis cheela* (35). First recorded, on Rakata, by Hoogerwerf in 1951, the serpent eagle was possibly seen on Rakata in 1985 and may have been heard there and on Panjang in 1990 (but was definitely heard on Sertung in 1990). On a conservative assessment we discount its presence on Rakata, but it is possible that it has persisted since 1951 (see above).

Black-naped fruit dove, *Ptilinopus melanospila* (37). First discovered, feeding on figs on Rakata, in 1951 by Hoogerwerf (1953). Netted on Rakata in 1984 and seen in 1985 (Zann *et al.*, 1990b), the fruit dove was again recorded there (as very common) in 1990 and (as common) in 1992.

Olive-winged bulbul, *Pycnonotus plumosus* (39). Seen in coastal vegetation on the island in 1984 by Geoffrey Davison (Zann *et al.*, 1990b) this bulbul was then assessed as present but rare (Thornton *et al.*, 1990b). The species has not been recorded since, however, and is now assessed as absent from Rakata.

Oriental hobby, *Falco severus* (40). In November 1952 Hoogerwerf (1953) saw one individual on Sertung in *Casuarina* heavily damaged by Anak Krakatau's October eruption. One individual was seen around Rakata's summit in 1982 (Alain Compost, pers. comm.) but the species has been seen since only on Anak Krakatau (on our 1986 expedition and in 1989 by Haag & Bush). It is now counted as a straggler on Rakata in 1982, and is assessed as not being resident on the island.

Barn owl, *Tyto alba* (43). See also below.

Chestnut-capped thrush, *Zoothera interpres* (45). An individual of this furtive species was netted on Rakata in 1984 and one was seen on Anak Krakatau during our 1986 expedition (Zann *et al.*, 1990b). The species was heard on Rakata several times in 1990 and classed as frequent. (Simon Cook netted four birds on Sertung in 1992).

Changeable hawk-eagle, *Spizaetus cirrhatus* (47). This species was seen once on Anak Krakatau in 1989 by Haag & Bush (1990), who believed the individual may not have been resident. We saw the hawk-eagle on Sertung and heard it on Panjang and Rakata in 1990. We both saw and heard it on Panjang and Rakata in 1992, and large inactive nests in tall trees, built on the lowest branches, near the trunk, were seen on both islands. The species is known to build more than one nest, however, using only one for breeding, so that these may have been nests of a single pair. A recent immigrant to the archipelago, the species is counted as present on Rakata.

Peregrine falcon, *Falco peregrinus* (49). One individual was seen on 6 July 1992 flying out over coastal vegetation at Owl Bay, Rakata, towards Panjang. Peregrines have otherwise only been recorded from Anak Krakatau. It is, however, likely that their home range includes all four islands, their 'core area' being on Anak Krakatau. In 1990 one was seen and followed with a theodolite telescope, flying from Anak Krakatau over the sea towards Sertung. In the absence of further evidence we do not count the peregrine as resident on Rakata.

Grey-cheeked green pigeon, *Treron griseicauda* (54). One male, seen on 6 July 1992 on Rakata by van Balen was probably of this species which occurs in Java and southern Sumatra, but may have been *Treron curvirostra*; it is difficult to distinguish between these species in the field. We do not count this single record on assessment; the status of the species may be clarified by future surveys.

Species listed as present on Rakata 1983–1989 by Bush & Whittaker but not recorded by our group

Honey buzzard, *Pernis ptilorhynchus* (48). This species was recorded on Rakata in August 1989 by Haag & Bush (1990), who regarded the individual as possibly being a migrant. MacKinnon (1988) states that the resident form is confined to western Java and a short-crested race visits Java in the winter; this sighting may thus have been of the resident form. For the present we assess the honey buzzard as being absent from Rakata.

Species assessed as present on Rakata 1984–86 by Thornton *et al.* (1990b), now believed to have been absent or adventive.

Large-billed crow, *Corvus macrorhynchos* (4). Hoogerwerf (1953) failed to find the species on Rakata during 9 days in 1951, but saw several on Sertung. The species was seen on Rakata in 1983, however, by Bush & Newsome (1986) but not on that island since. We earlier regarded the crow as probably having become extinct on the archipelago between 1984 and 1985 (Thornton *et al.*, 1990b; Zann *et al.*, 1990a,b). It was not recorded in the surveys of 1985, 1986 (which chiefly concerned Anak Krakatau), 1989, 1990 or 1992. We now count this crow as an extinction.

Savanna nightjar, *Caprimulgus affinis* (7). The Rakata record of this species since 1932–34 is of the call of a single individual in the beach area of Zwarte Hoek in 1985. A single individual was also heard in 1985 on each of Panjang (Bat Cave Beach) and the Sertung spit. Although regularly recorded from Rakata and Sertung from 1908 to 1933, the species was not found on the three older Krakatau islands in the 1951, 1983, 1984, 1989, 1990 or 1992 surveys. Hoogerwerf heard the nightjar in August 1952 along Anak Krakatau's beach, and it was present on that island throughout the 1980s and early 1990s. Until there is confirmation of the species' continued presence on Rakata we regard the individual heard at Zwarte Hoek as having been a stray from Anak Krakatau.

Brahminy kite, *Haliastur indus* (21). The single individual seen in 1984 is now thought to have been adventive; there have been no other records from Rakata since 1933.

White-tailed swift, *Rhaphidura leucopygialis* (36, Table 2). Earlier assessed as having been present on the island but missed in our 1984 and 1985 surveys, the species was not recorded in any of the subsequent surveys and we now believe it to have been absent in the 1980s also.

Oriental hobby, *Falco severus* (40). The single individual seen in 1982 by Alain Compost constitutes the only record from Rakata and the individual is now thought to have been adventive, possibly from Anak Krakatau. We believe it has now been replaced on the archipelago by the peregrine falcon.

Barn owl, *Tyto alba* (43). We do not exclude owls from our data. They may be seen at night, when they are also

TABLE 3. Resident land bird species of Rakata, 1908–1992, based on assessment of records on the basis of minimum turnover. Data from Table 2.

Intersurvey period	I		II		III		IV		V	
Survey dates		1908		1919–21		1932–34		1951		1983–93
Intersurvey interval (years)	25		13		13		17		41	
Actual number of species		15		27		28		29		31
Cumulative number of species		15		29		32		38		43
Gains	15		14		3		6		6*	
Losses	—		2		2		5		4	
I (immigration rate)†	0.60		1.08		0.23		0.35		0.15	
E (extinction rate)†	—		0.15		0.15		0.29		0.10	

*One immigrant, *Ducula bicolor*, is a well-known 'in-and-out' species that colonises for short periods opportunistically (Hoogerwerf, 1953).
†In species per year.

vocal, their roosts and nest sites are often indicated by droppings, and they produce characteristic pellets. The barn owl was first seen at Owl Bay, Rakata, in 1982 by Thornton (Zann *et al.*, 1990b) and in that year was recorded from Anak Krakatau by Ibkar Kramadibrata *et al.* (1986), whose survey did not include Rakata. It has now been shown to be breeding on Anak Krakatau (Rawlinson *et al.*, 1992). The species was not recorded on Rakata in 1989 by Haag & Bush, nor in our 1990 or 1992 surveys when no night work was done on Rakata. Owl Bay, however, has been a frequent camp site of a number of expeditions since 1982, and no further indication of the owl's presence has been noted. We thus do not count the owl as resident on Rakata.

COMMENTS

Bush & Whittaker (1991 p. 346) stated, 'by 1951 seven of the original thirteen resident land bird species had become extinct'. From our reading of their Table 2 this should have read 'six of the original thirteen'. They then noted that three of these were observed to have had declining populations in 1951, and two of the three were not recorded on Rakata between 1951 and 1983. There was no bird survey between 1951 and 1983, but one of the three, *Centropus bengalensis*, has not been recorded on Rakata in the 1983–1992 surveys. They continued 'the four remaining species from the 1908 avifauna all exhibit coastal distributions'. It is not clear whether this statement refers to the four still remaining on Rakata in 1983–89 according to their Table 2 (our data show seven species still on Rakata), or to the four that are left after the three with declining populations have been considered. Either way the following species would be included, which do not have coastal distributions: *Halcyon chloris*, seen and netted in *Neonauclea* forest, where it was common up to 250 m, as well as in *Barringtonia/Terminalia* habitat, and *Pycnonotus goiavier*, recorded in all habitats except the beach, including *Neonauclea-Ficus* forest above 400 m and the *Schefflera* scrub on Rakata's summit at about 770 m. *Oriolus chinensis*, which would be included if the first of the alternatives above is correct, was recorded in *Barringtonia/Terminalia* habitat, *Neonauclea* forest below 400 m, and in *Neonauclea-Ficus* forest up to 640 m.

On page 348 Bush & Whittaker (1991) stated, 'other than [extinctions of 1908 species] only three species have gone extinct on Rakata since 1951 (Table 2)'. According to their Table 2, this should read 'five species'. They continued, 'of the nine bird species recorded since 1934 on Rakata but not found on that island in the 1980s, one was documented from Sertung and a further five species from Anak Krakatau'. Inspection of their Table 2 shows that the number of species quoted should read eight, one, and three, respectively. Using our data, the figures become four, three, and two respectively.

RESULTS

Thornton *et al.* (1990a) provided a colonization curve and tables of immigration and extinction rates for Rakata, based on work up to 1986, showing values based on records alone and on an assessment of these. In Table 3 and Fig. 1 we bring our 1986 data for Rakata up to date in the light of the 1989, 1990 and 1992 surveys (see above), grouping the 1983–1992 survey data. In Fig. 2 we plot the Rakata immigration and extinction rates, (based on minimum turnover) against survey dates for comparison with Bush & Whittaker's Fig. 7, and in Fig. 3 these rates are plotted against average number of species present in the intersurvey intervals.

Species numbers, rather than having fallen by 20% since 1951 (Bush & Whittaker, 1991, Table 2, Fig. 8), have risen by between 7% and 44%, depending on whether all records are accepted at face value (from twenty-seven to thirty eight species, 44%), or subjected to assessment and the principle of minimum turnover applied (from twenty-nine to thirty-one species, 7%) (our Table 2). The colonization curve (Fig. 1) does not fall, as depicted by Bush & Whittaker, rather the rate of increase of species numbers has declined markedly since the early 1920s (the time of forest formation) and the curve has flattened considerably.

Immigration rate (Fig. 2) rose in intersurvey period II, then declined in period III to rise slightly in period IV and then drop to its lowest value in period V. The 1931 survey was made without an ornithologist and immigration rate based on this survey (i.e. in period III) would be underestimated if species had been missed. Extinction rate based on records would be overestimated if species were missed; extinction rate based on minimum turnover would be little affected. The general trend of Figs 2 and 3 is an initial increase in the immigration rate followed by a decline to a low value approaching the extinction rate; there is little evidence of non-monotonic change apart from the early rise at the time of forest formation (Fig. 2). Extinction rates are low throughout and substantially the same for each intersurvey period.

DISCUSSION

Bush & Whittaker, believing that bird numbers had fallen since 1951, and noting that immigration and extinction rate time curves for birds and butterflies were not monotonic, asserted, 'The change from grasslands to forest during the 1920s appears to have been accompanied by non-monotonic changes in immigration and extinction rates. *This observation alone* would cast doubt on the ability of equilibrium theory to provide the basis for a predictive model of recolonization (Bush & Whittaker, 1991 p 354) (our stress).

MacArthur & Wilson (1967) suggested that theoretically the curve for island immigration rate (plotted against number of species present) should fall smoothly, and that for extinction rate should rise smoothly, to meet at an equilibrium condition. They acknowledged, however (pp. 50, 51), that 'we still know very little about the precise shape of the extinction and immigration curves, so that few predictions can be made'. For the plants, birds and butterflies of the Krakataus we now know something of the shape of these curves (Fig. 1–15).

The numbers of immigrants and extinctions are low, and thus rates are highly sensitive to decisions as to the true immigrant status or extinction of species, particularly when intersurvey intervals are short (see Table 1). These decisions have been made here after careful assessment, but a difference of only one or two species may alter the curves considerably. Several general trends, however, are apparent.

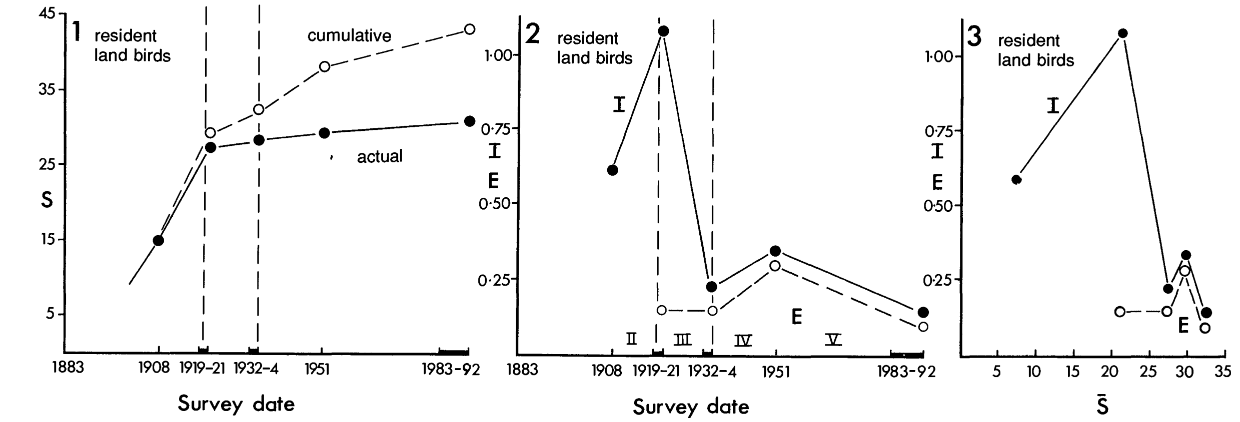
On the basis of minimum turnover, the immigration rate of vascular plants to Rakata (data from Whittaker, Bush & Richards, 1989) (Fig. 14) fell initially (c.f. Bush & Whittaker, 1991, Fig. 3b), as expected by theory, but then rose, to fall again. The initial fall is greater for pteridophytes (Fig. 8) than for spermatophytes (Fig. 11 c.f. Bush & Whittaker's Fig. 3a) and the subsequent rise in the rate of pteridophytes lags behind that of spermatophytes by one survey, occurring not before, but during, forest formation. The shapes of the immigration rate curves suggest that in each we may be seeing the resultant of two curves, that of an early phase of colonization, in which the pool of potential pioneer colonizing species quickly became depleted, and that of a subsequent phase of secondary colonizers, the potential pool of which was depleted more slowly. The species–time curve for pteridophytes (Fig. 7) is also of a shape suggesting the combination of two phases of increase in species numbers, one before, one during, forest formation. This is not seen in the curve for spermatophytes (Fig. 10) nor therefore in the curve for vascular plants (Fig. 13), perhaps because the spermatophyte phases are actually less distinct. The difference in the time of the peaks of the two immigration rate curves suggests that the second phase was rather later in the case of pteridophytes than spermatophytes; possibly the beginning of forest formation, as a result of spermatophyte succession, with its attendant changes in microclimates, was a prerequisite for the second wave of pteridophytes, which, unlike the first wave, was composed largely of shade species.

The two groups of animals for which there is good data, involving a substantial number of species over time, are resident land birds and butterflies. We use the data of Table 1 of Bush & Whittaker (1991) for butterflies, omitting *Rapala iarbus* and *Anapheis java* as being migrants, and applying minimum turnover assumptions, as for birds and plants. Both birds and butterflies show a somewhat different pattern of changes in immigration and extinction rates (Figs 2 and 5 respectively) from that of spermatophytes (Fig. 11) even accounting for the lack of early animal surveys. Immigration rate of birds and butterflies rose to a maximum (Figs 2, 3, 5, 6), in contradiction of the basic theory, the rise (as for pteridophytes, Fig. 8) coinciding with forest formation, followed by the expected decline. There were no early zoological surveys, the first being 25 years after the 1883 event, so that if there had been a 'pioneer' effect, as suggested above for plants, it would not have been monitored. The peaks of butterfly, bird, and pteridophyte immigration rates lag behind that of spermatophytes (Fig. 11) by one survey, as may be expected in a successional situation, suggesting the influence of changes in spermatophytes on other colonizing groups.

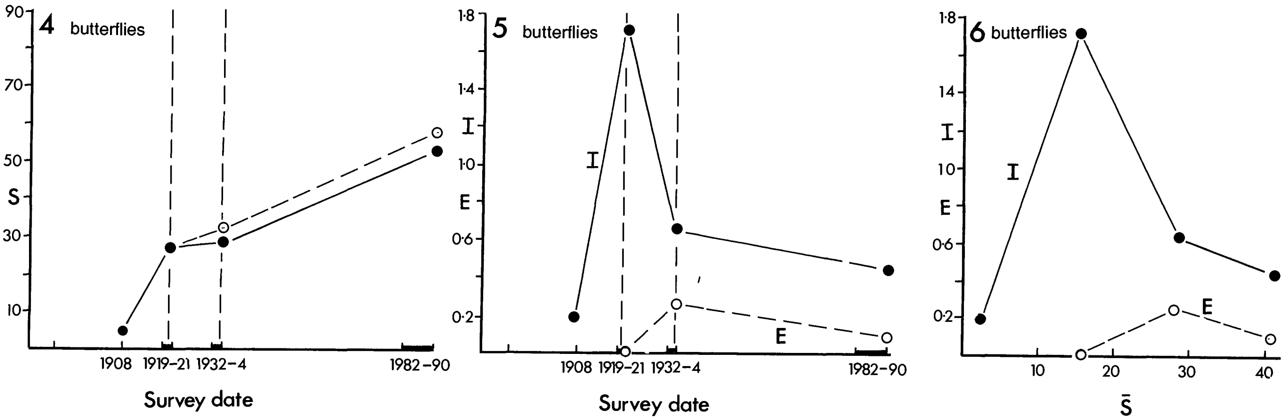
Extinction rates, according to theory, should be low initially, then gradually rise to meet falling immigration rates at equilibrium (MacArthur and Wilson, 1967). Whilst the extinction rates of plants rose, they did not do so gradually; the rise for both pteridophytes (Figs 8, 9) and spermatophytes (Figs 11, 12) was most marked at about the time of forest closure, when they at least doubled. The extinction curve of butterflies (Fig. 5) is similar to that of pteridophytes (Fig. 8), although there are fewer data points. The extinction rate of birds (Figs 2, 3) has remained fairly constant at about one species every 7 years; that of butterflies (Figs 5, 6) ranges from zero to one every 2 years. The butterfly species accumulation curve (Fig. 4) provides no evidence of an approach to equilibrium by this group.

Thus, on the evidence, there are the following differences from the gradual mutual approach of the immigration rate and extinction rate curves that would be expected from the basic theory. For plants (no equivalent data are available for animals) there is some indication of an early pioneer phase of colonization followed by a phase of secondary colonizers. Immigration rate of spermatophytes rose to a peak prior to forest formation (Fig. 11), and that of pteridophytes peaked during this period (Fig. 8) then fell; there was a marked rise in extinction rate of both groups at about the time of forest canopy closure. The early pioneer phase of colonization by birds and butterflies, if there was a distinct one, was not monitored, but their immigration rates, which were first measured later than those of plants, rose (Figs 2 and 5 respectively) with that of pteridophytes, following the spermatophyte immigration rate peak as forest formation began, then declined.

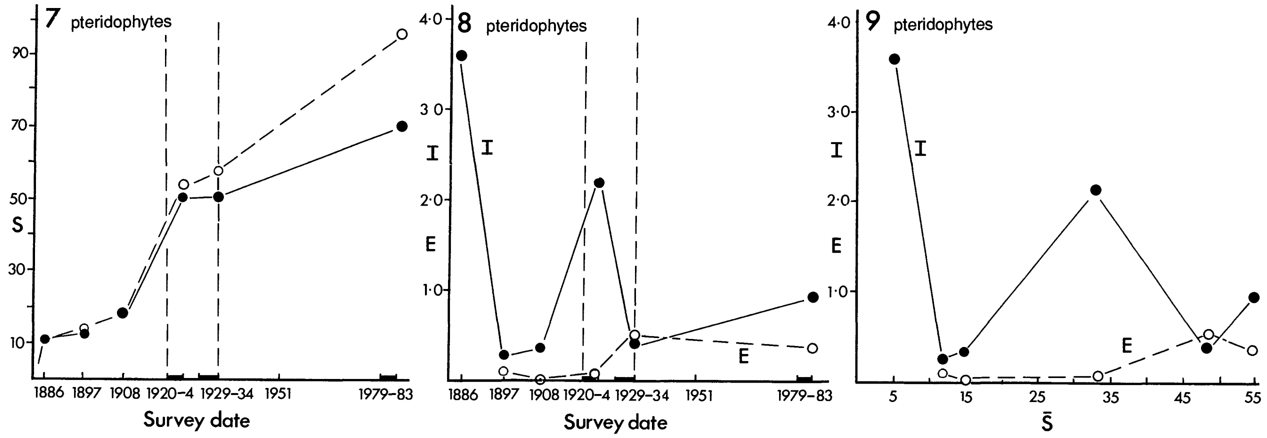
Contrary to the statement of Bush & Whittaker (1991, p. 346) the immigration rate of butterfly species has not risen. Using the data of their Table 1 and applying minimum turnover technique, between the 1919/21 and 1933 surveys



FIGS 1–3. Resident land birds, Rakata, 1908–1992: Fig. 1, species (S) accumulation (colonization) curves; Fig. 2, change with time of immigration (I) and extinction (E) rates, as species per year for intersurvey periods II–V; Fig. 3, change of I and E with average number of species (S) in intersurvey intervals. Vertical dashed lines represent period of forest formation; dark horizontal bars indicate closely successful surveys that are integrated; open circles in Fig. 1 indicate cumulative, closed circles actual, number of species. Data from Table 3.

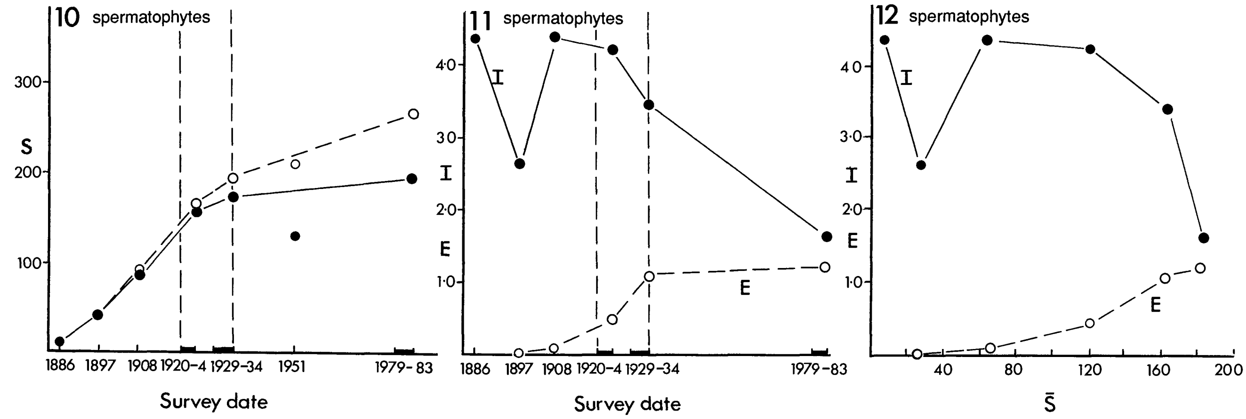


FIGS 4–6. Butterflies, Rakata, 1908–1990: Fig. 4, species accumulation curves; Fig. 5, changes of immigration and extinction rates with time; Fig. 6, changes of immigration and extinction rates with average number of species present. Symbols and abbreviations as Figs 1–3. Data from Bush & Whittaker (1991) Table 1 (see text).

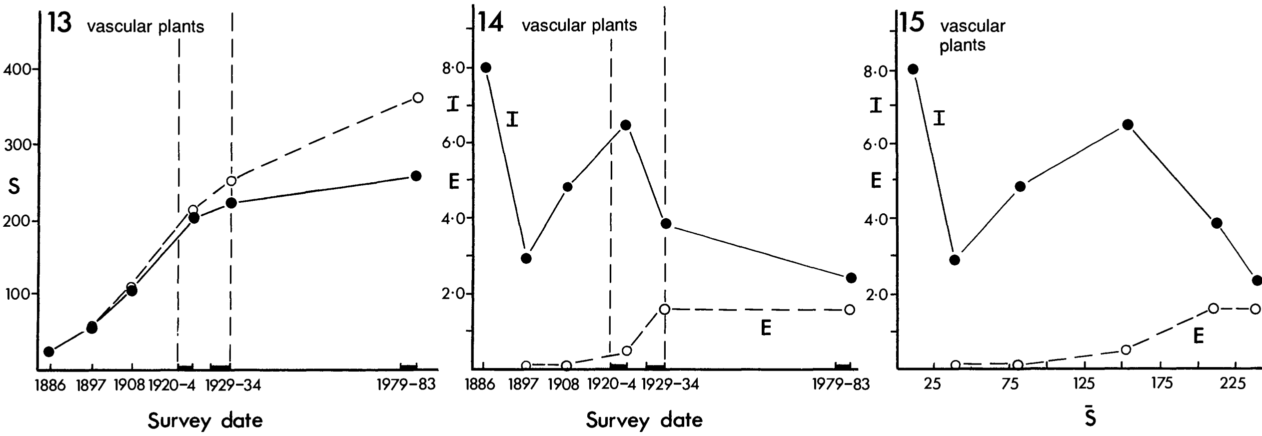


FIGS 7–9. Pteridophytes, Rakata, 1886–1983: Fig. 7, species accumulation curves; Fig. 8, changes of immigration and extinction rates with time; Fig. 9, changes of immigration and extinction rates with average number of species present. Symbols and abbreviations as Figs 1–3. Data from Whittaker *et al.* (1989).

eight species immigrated (immigration rate 0.7 spp/y) and between 1933 and 1982/89, twenty-six species (0.5 spp/y). Moreover, their Fig. 5 also shows a decline, not an increase, in immigration rate over the last intersurvey period. Bush & Whittaker use 'the present rising rate of immigration' of butterfly species, together with falling extinction



FIGS 10–12. Spermatophytes, Rakata, 1886–1983: Fig. 10, species accumulation curves; Fig. 11, changes of immigration and extinction rates with time; Fig. 12, changes of immigration and extinction rates with average number of species present. Symbols and abbreviations as Figs 1–3. Data from Whittaker *et al.* (1989).



FIGS 13–15. Vascular plants, Rakata, 1886–1983: Fig. 13, species accumulation curves; Fig. 14, changes of immigration and extinction rates with time; Fig. 15, changes of immigration and extinction rates with average number of species present. Symbols and abbreviations as Figs 1–3. Data from Whittaker *et al.* (1989).

rates, as a reason to 'deny the proximity of a MacArthur & Wilson-style equilibrium for butterflies'. There is certainly no evidence of an approach to such an equilibrium for butterfly species, but a 'present rising rate of immigration' cannot be used as support for this assertion. Bush & Whittaker also stated (p. 345), 'the immigration and extinction curves . . . intersect by the end of the 1920s (Fig. 5)'. They do so on their Fig. 5 only if one assumes that the records accurately reflect extinctions; their minimum turnover curves (as our Fig. 5) do not intersect.

Extinction rates of birds and butterflies have not risen consistently to approach the falling immigration rates (as they have in spermatophytes). Although a peak is seen in the extinction rate curve of butterflies (Figs 5, 6) which corresponds to about the time of canopy closure, there is no such peak in the case of birds (Figs 2, 3), possibly reflecting the closer ecological ties between butterflies and plants, and their greater difficulty, therefore, in persisting in a rapidly changing floristic environment.

We see nothing in the data to warrant the general conclusion that the model has been found wanting. We believe it is possible that we are observing the results of a number of superposed, partially overlapping, colonization series, corresponding to major successional changes in the island's environment (e.g. development of grassland, forest formation, forest canopy closure), changes which may affect the various subsets of the biota differentially. The environment of the target of colonization, the islands, is constantly changing (partly as a result of colonization), and thus so are the *effective* available pools of groups (e.g. birds, pteridophytes) of potential colonizers. The changes in these pools will vary in degree between different groups of organisms, and it is not surprising that the basic theoretical curves are not clearly or easily evident.

MacArthur & Wilson pointed out that succession (forest formation) may be important in the early stages of colonization by Krakatau birds, and may explain the 'more rapid increase [of bird species] between 1908 and 1920 than between 1883 and 1908'. They acknowledged that the rate curves may be non-monotonic in situations where succession is important in the early stages of colonization and involves different trophic levels. They suggested, for example, that the extinction curve may initially decline where the heterogeneity of the area permits earlier colonists to persist although new ones have become established. On the evidence of our data, this does not seem to have occurred on the Krakataus. Rather, extinction rate of birds has not risen, and those of butterflies and plants rose at about the time of forest closure. Immigration rate, however, after an initial pioneer phase had run its course, has fallen; were it to fall to the same level as extinction rate and be maintained there, there would be equilibrium of course, in spite of the lack of a rise in extinction rate.

On neither Rakata nor the Krakataus is an equilibrium of butterfly species numbers evident (Yukawa, 1984; New *et al.*, 1988, Thornton & New, 1988, Thornton *et al.*, 1990a). Although the data for birds and plants may be interpreted, if one follows the MacArthur–Wilson model, as indicating that equilibria are being approached (flattening of the species-time curves, approach to proximity of the immigration and extinction rate curves) there is no evidence that this is so for butterflies, whose numbers are still rising fairly steeply. We do not believe, however, that the theory implies that equilibrium would be achieved, or approached, by all segments of a biota simultaneously (e.g. Thornton *et al.*, 1990a; Thornton, 1991). The dependence on the prior presence of particular species of plants for successful colonization is probably more rigorous for butterflies than it is for birds, few of which have such strict, specific plant requirements, and, again if one follows the model, butterfly species numbers may therefore be expected to equilibrate (if they do) more slowly than those of birds.

Bush & Whittaker (1991, p. 352) argued that because the majority (actually 59%) of butterfly species first recorded in 1979 utilize food plants which have been present since 1908–1920, 'we may reject the hypothesis that the majority of butterfly species have had to await the arrival of food-plants'. This test, however, which was thought to prove that butterflies were 'genuinely rare arrivals', loses much of its force when one considers that between 1933 and 1979 there was no butterfly survey. Species first recorded in 1979 could have become established a decade or so after their food plants and then had to await their discovery some 46 years later to become new records.

Bush & Whittaker (1991) stated, 'simple notions of equilibrium are thus shown to be inadequate in such complex ecosystems'. Certainly the MacArthur–Wilson theory in its basic form cannot fully explain the Krakatau data, but this was acknowledged by its authors. It is unlikely, in its pure form, to fully explain the build-up of a complex biota either where much of the turnover is successional or where there have been local disturbances setting back or deflecting succession. Both of these are relevant to the Krakatau situation as Whittaker and his colleagues have so ably demonstrated for plants, although local disturbances have been least on Rakata. In our previous publications (e.g. Thornton *et al.* 1990a,b) we have used the theoretical model as a framework against which to set the observed data, and have found this approach to have heuristic value, as, we believe, was MacArthur & Wilson's main intention. For example, clear differences from the theoretical curves may be instructive in pointing up departures from the simple model (which treats all species as interchangeable), such as the suggestion of two overlapping colonization sequences from the colonization curve of pteridophytes, and the rise in their immigration rate, along with birds and butterflies [and aculeate Hymenoptera (Yamane, Abe & Yukawa, 1992)] at the time of forest formation. Comparison of the curves of different components of the biota may highlight different responses to the changing environment. The near equilibration, alone of the plant dispersal-groups, of the sea-dispersed component (Whittaker *et al.*, 1992), for which Bush & Whittaker (1991) have offered an explanation, is one example. Another is the much slower approach to equilibrium by butterfly species numbers than by those of birds, which we have suggested results from the different nature of the ecological links with plants of these animal groups.

The generally accepted 'hierarchical' dependences by certain groups for their successful establishment on the presence of other groups of organisms, or on particular physical changes that are concomitants of succession, dependences that were well recognized by Treub (1888), Docters van Leeuwen (1936), Dammerman (1948), and by MacArthur & Wilson (1967), will of course tend to complicate the basic theoretical curves but in our view do not invalidate the model's usefulness.

Although, as Bush & Whittaker suggest, a millenial time scale may well be necessary for any equilibration of the forest tree component of the biota, and, therefore, for that of the total ecosystem, it should not be assumed that all components would equilibrate at the same rate as forest trees. The strengths of community links with forest trees and the degree of congruence between changes in other groups of organisms and those of the forest trees may be expected to vary among the various components of the biota. Some may have close, direct ecological ties with trees; for others there may be a number of intervening links and a much more indirect relationship, resulting in a greater degree of independence from the slow changes in forest tree component species and dominance patterns which Bush & Whittaker predict. At the extremes, a eurytopic scavenger-omnivore such as *Varanus salvator* is unlikely to respond to such changes as faithfully as, for example, stenophagous species of Lepidoptera, or bats which have highly specialized roosting requirements such as species of *Tylonycteris*. As already noted here for birds and butterflies, and elsewhere (Thornton *et al.*, 1990a, Thornton, 1991) for other groups, the evidence suggests that only some components of the biota are moving towards asymptotes, and these are doing so at different rates. Such a differential pattern is likely to continue as forest succession and maturation slowly proceeds.

Thus, although Bush & Whittaker's criticism of the applicability of the dynamic equilibrium island biogeographic approach to the colonization of forest trees may be valid, in our view extrapolation to include all biotic components, and extension to all specific localities, all complex forest communities, and even the lowland tropics generally, is not warranted by the evidence so far accrued on the Krakataus.

The MacArthur–Wilson theory regards the constancy of island species numbers as being due to a dynamic balance between immigration and extinction, the identity of some of the species present changing over time, but the total number fluctuating about an equilibrium that is characteristic for a particular island. Bush & Whittaker (1991) advocate a move to the ideas of Lack (1976), by which constant species numbers are maintained about a figure which is determined by the fixed number of available niches on the island, rather than by the resultant of a dynamic equilibrium. On this model potential colonists are excluded by incumbent species and extinctions are few; there is little or no turnover on an island once it has received its complement of species, apart from that of transients, whose status as components of the biota are arguable.

Both Whittaker's group and ours agree that on the Krakataus turnover in plant and animal species has thus far been largely successional, with a considerable core of species, best identified in plants by Bush & Whittaker (e.g. 1991) which has changed little in numbers or identity over a long period. This suggests a pattern similar to that found in the experimental aquatic microcosms of Dickerson & Robinson (1985), but it must be noted that as yet on the Krakataus we have no equilibrium to test; we are still monitoring the process of reassembly. Nevertheless, many of the findings of experimental community ecologists, such as Robinson & Dickerson (1987) and Drake (Drake, 1991; Drake *et al.*, 1993), could be studied-profitably in the context of the Krakatau case. We who study 'natural experiments' may learn much from those who assemble, and can thus manipulate, their own microcosms in the laboratory.

ACKNOWLEDGMENTS

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